Inferring complexity from Web data

Discussions

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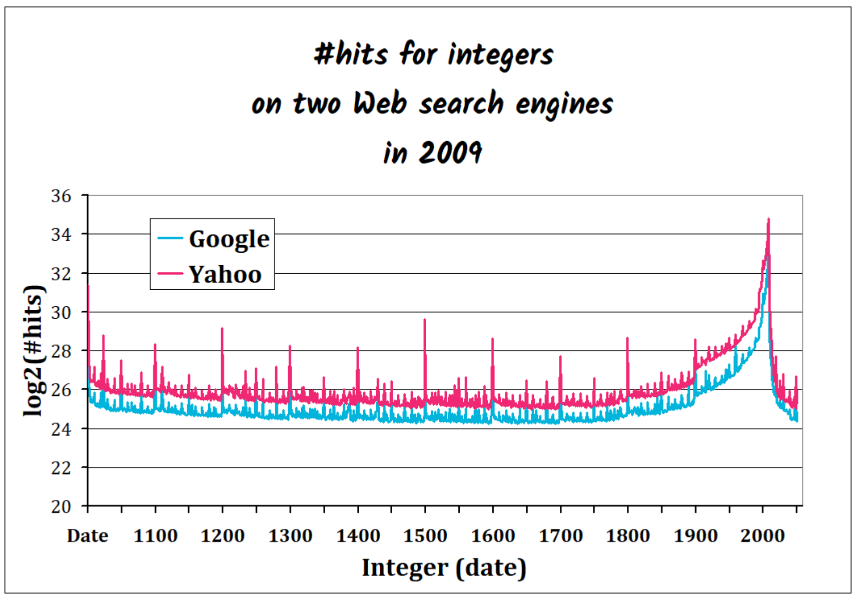
Problem: Is there a link between complexity and Web frequency?

As we saw, some integers, such as round numbers, are simpler than others. Does it affect their frequency on the Web? The picture below represents the number of hits (in 2009) *declared* by Yahoo and Google for requests consisting of numbers between 1000 and 2050 (the correlation for both search engines is 0.98; note the logarithmic vertical scale).



One can unambiguously see that larger frequencies in the Web correspond to "simple" numbers:

- Multiples of 100
- 1024
- Multiples of 50
- Recent dates

Complexity and frequency

problems,

average address length is minimal when

the address of a word w is equal to the

logarithm of its inverse frequency (up

to a constant):

When a word like “selfie” becomes

trendy,

its frequency increases in

language.

Question: *stallion* is more complex (and less frequent) than *horse*. What about *quadruped*?

Answer: *quadruped* is indeed less frequent than *horse*. The point is that *quadruped* is more complex a concept than *horse*, even if a horse is a quadruped with additional characteristics (just ask children about horses and quadrupeds).

Some authors suggested the possibility of inverting the relation between complexity and frequency. The frequency of words in texts can be assessed using a search engine. Though it is difficult to infer the real number of pages indexed by search engines, it seems that dividing Google's declared number of hits by 2×10^{12} gives a reasonable estimate of actual frequency. For instance, the word "spoon" can be considered to occur with a frequency of 0.02%, whereas the frequency of "book" is 0.87%.

If a word w occurs with a frequency f, we get an **estimate of the complexity** $K(w)$ of the corresponding concept through the formula:

$$K(w) \leq \log(1/f).$$